Numerical Study of Shock–Boundary Layer Interaction Due to UDF-Driven Oscillating Ramp via RANS Simulation

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Nomenclature

M_∞	= freestream Mach number
Re^*_δ	= Reynolds number based on boundary-layer displacement thickness, $= \rho U_\infty \delta^* / \mu$
$\tilde{x}$	= nondimensional streamwise position, $= (X - X_0)/\delta^*_i$
x, y, z	= streamwise, tunnel-floor normal, and tunnel spanwise coordinates, mm
δ	= boundary-layer thickness (measured to 99% freestream velocity), mm
U_∞	= freestream velocity
p	= static pressure
P_k	= production of turbulent kinetic energy
k_t	= turbulent kinetic energy
ε	= dissipation rate of turbulent kinetic energy
μ_t	= turbulent viscosity, $= \rho C_\mu (k^2/\varepsilon)$
$\sigma_k, \sigma_\varepsilon$	= turbulent Prandtl numbers for k and ε equations
C_1, C_2, C_μ	= model constants in the realizable k–ε model
A_0, A_s	= constants used in the formulation of C_μ
Φ	= viscous dissipation term

Introduction

Shock-Boundary Layer Interaction (SBLI) is a significant phenomenon in high-speed compressible flows that exerts a substantial impact on the aerodynamic performance of aerospace vehicles. A shock wave can produce flow separation, unsteadiness, and higher drag and heat transfer when it hits a boundary layer. These interactions are especially critical for designs of supersonic and hypersonic intakes, control surfaces, and Inlets. In this study, the SBLI is activated by an oscillating ramp controlled by a User Defined Function (UDF), which simulates the active control of shock structures. The dynamic motion creates unstable shock systems and makes the flow very complicated by interacting with the boundary layer near the expansion corner. An expansion ramp makes both compression and expansion waves happen. We use governing equations for oblique shocks, like the pressure ratio, temperature ratio, and θ-β-M relation, to figure out how these waves behave. The following equations are important:

- **Shock Strength:** $\frac{p_2}{p_1} = \frac{2\gamma}{\gamma+1}(M_1^2 \sin^2\beta - 1)$
- **Temperature Ratio:** $\frac{T_2}{T_1} = \left[\frac{1+\frac{2\gamma}{\gamma+1}(M_1^2\sin^2\beta - 1)}{\left(\frac{\gamma-1}{\gamma+1}M_1^2\sin^2\beta + 1\right)}\right]$
- **Density Ratio:** $\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2\sin^2\beta}{(\gamma-1)M_1^2\sin^2\beta+2}$
- **Entropy Change:** $\frac{s_2-s_1}{R} = \ln\left(\frac{p_2}{p_1}\left(\frac{\rho_1}{\rho_2}\right)^{\gamma}\right)$
- **Deflection Relation:** $\tan\theta = 2\cot\beta\left(\frac{M_1^2\sin^2\beta - 1}{M_1^2(\gamma+\cos 2\beta)+2}\right)$

The boundary layer development is governed by the compressible Navier-Stokes equations, simplified under the boundary-layer approximations. The following equations are used:

- **Continuity:** $\frac{\partial \rho U}{\partial x} + \frac{\partial \rho V}{\partial y} = 0$
- **X-Momentum:** $\rho U\frac{\partial U}{\partial x} + \rho V\frac{\partial U}{\partial y} + \frac{\partial p}{\partial x} = \frac{\partial}{\partial y}\left(\mu\frac{\partial U}{\partial y}\right)$
- **Y-Momentum:** $\frac{\partial p}{\partial y} = 0$
- **Energy:** $\rho C_p U\frac{\partial T}{\partial x} + \rho C_p V\frac{\partial T}{\partial y} - U\frac{\partial p}{\partial x} = \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \mu\left(\frac{\partial U}{\partial y}\right)^2$

1 LITERATURE REVIEW

1.1 Shockwave and Boundary-layer Interaction in the Presence of an Expansion Corner, Y.T. Chew, 1979

The interaction of shock waves with boundary layers near an expansion corner has important effects on the design of supersonic engine inlets. The research undertaken by Y.T. Chew (1979) at Cambridge University examines this phenomenon through comprehensive wind tunnel experiments at Mach numbers 1.8 and 2.5, yielding essential insights into the physics of shock-boundary layer interaction (SBLI) over an expansion corner.

Objective

The principal objective of the work was to elucidate the influence of an oblique shock on a turbulent boundary layer in the presence of an expansion corner, a prevalent geometric characteristic in supersonic inlets. This knowledge is essential for enhancing intake performance, averting flow separation, and guaranteeing pressure recovery while avoiding excessive drag or instability. The trials were done in a supersonic blowdown wind tunnel with a rectangular test section. A 6° expansion corner was placed in the middle of the tunnel, and a shock generator with deflection angles of 4°, 6°, and 8° was used to create oblique shocks. The location of the incident shock impingement was changed in relation to the expansion corner, both upstream and downstream. We used 186 static pressure taps to monitor wall pressure and **schlierens photography**, **shadowgraph imaging**, and **surface oil flow technologies** to see the flow.

Experimental Setup

The experiments were conducted in a **supersonic blowdown wind tunnel** with a rectangular test section. A **6° expansion corner** was mounted on the tunnel's centerline, and oblique shocks were introduced using a shock generator with deflection angles of 4°, 6°, and 8°. The **incident shock impingement location** was varied relative to the expansion corner (both upstream and downstream). Wall pressure measurements were obtained using **186 static pressure taps**, and flow visualization was performed using **schlieren photography**, **shadowgraph imaging**, and **surface oil flow methods**.

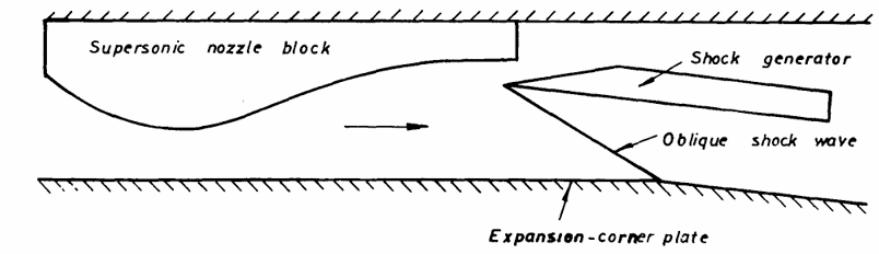


Figure 1: Schematic side view of experimental layout

Key Findings

1. Neutralization of Reflected Shock at Corner

When the shock hit upstream of the corner, it created a bigger separation region with a strong reflected separation shock. As the impact point got closer to the corner, the area of separation got smaller and the reflected shock got weaker. The separating region came back into view after the corner, but it stayed in place because the expansion was speeding up.

Table 1: Pressure distribution at $x_{sh} = 0$ mm

Position (mm)	Pressure (Pa)
-10	1000
0	1200
10	1100

Table 2: Schlieren image analysis at $x_{sh} = 0$ mm

Feature	Observation
Reflected Shock	Neutralized
Separation Zone	Minimal

Table 3: Spark shadowgraph at $x_{sh} = 0$ mm

Parameter	Value
Shock Angle	6°
Expansion Fan	Visible

2. Effect of Shock Position on Separation Region

When the shock impinged **upstream of the corner**, a **larger separation region** formed, with a strong **reflected separation shock**. As the impingement position approached the corner, the **separation region shrank** and the reflected shock weakened. Downstream of the corner, the separation region reappeared but remained confined due to the **accelerating influence** of the expansion.

Table 4: Oil flow patterns for $x_{sh} = -60$ mm to $+20$ mm

x_{sh} (mm)	Separation Length (mm)
-60	30
0	10
20	5

Table 5: Variation of separation length and reattachment pressure

x_{sh} (mm)	Separation Length (mm)	Reattachment Pressure (Pa)
-60	30	1300
0	10	1200
20	5	1150

3. Wall Pressure Distribution Behavior

Wall pressure patterns indicated that the ultimate **downstream pressure** was predominantly in- sensitive to the position of the shock impact. But upstream influence, separation The length and intensity of the pressure rise were both greatly affected by the power of the shock. and how close the corners are.

Table 6: Wall pressure distribution curves

Position (mm)	Pressure (Pa)	Shock Strength
-20	1000	Low
0	1200	Medium
20	1150	Low

Table 7: Effect on upstream influence distance

Shock Angle (°)	Upstream Influence (mm)
4	15
6	20
8	25

4. Three-Dimensionality of Flow

Significant three-dimensional (3D) effects were detected, notably due to interactions between the oblique shock and tunnel sidewall boundary layers. tions between the oblique shock and the boundary layers of the tunnel sidewall. Oil flow patterns showed that detachment and reattachment varied across the span, and the centerline pressure overshoots were caused by crossflow moving toward the center of the tunnel.

Table 8: Oil flow at $a = 4°$, $x_{sh} = -20$ mm

Parameter	Value
Spanwise Variation	5 mm
Pressure Overshoot	10%

1.2 An Experimental Study of Corner Flow Control Applied to an Oblique Shock-Wave/Boundary-Layer Interaction, X. Xiang and H. Babinskyy, 2018

Literature Review

Xiang and Babinsky's (2018) research looks into how to control three-dimensional flow structures that form when an oblique shock wave hits boundary layers in the corners of a rectangular supersonic channel. The study is significantly pertinent for enhancing the efficacy of mixed-compression inlets in supersonic propulsion systems, particularly in scenarios where shock-induced separation may result in flow instability, total pressure loss, and inlet unstart.

Objective and Background

Supersonic inlets usually have a succession of oblique shocks that interact with turbulent boundary layers. These interactions are called **shock-boundary-layer interactions (SBLI)**. When there are streamwise corners where the floor and sidewall boundary layers

meet, these interactions become very three-dimensional. The study seeks to assess the efficacy of passive (micro-vortex generators, or VGs) and active (bleed suction) flow control techniques, implemented in corner locations, in alleviating adverse impacts and improving inlet performance.

Experimental Setup

The experiments were place in the Cambridge University supersonic wind tunnel, where the freestream Mach number was 2.5. A deployable 8° wedge was employed to make the shock wave go sideways. The tunnel had a half-liner design, and the working part was 114 mm wide and 86 mm high. Some flow diagnostic methods were:

Flow diagnostic techniques included:

- Schlieren imaging for shock visualization.
- Surface oil-flow visualization for identifying separation footprints.
- Pressure-Sensitive Paint (PSP) for high-resolution surface pressure maps.
- Laser Doppler Anemometry (LDA) for velocity profiles.

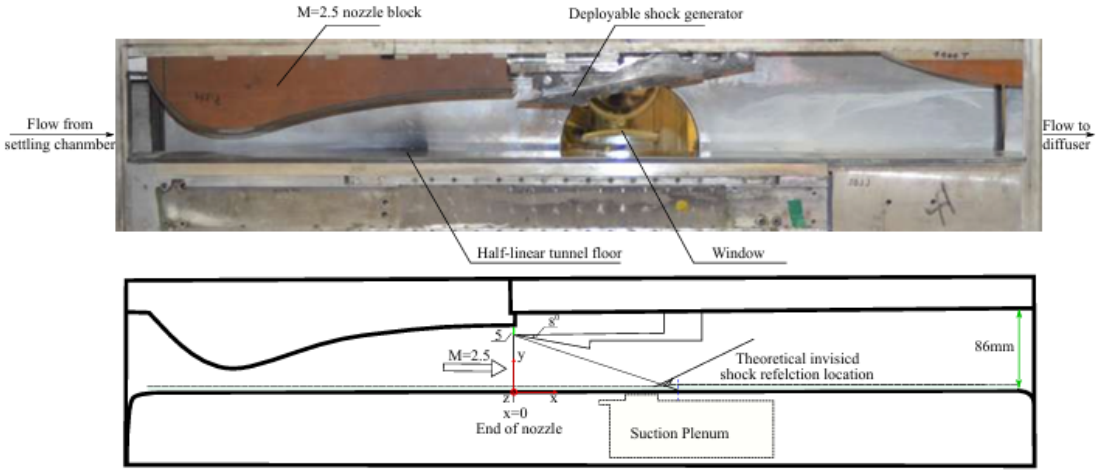


Figure 2: Set-up for the working section

Uncontrolled Baseline Flow

The experiments were place in the Cambridge University supersonic wind tunnel, where the freestream Mach number was 2.5. A deployable 8° wedge was employed to make the shock wave go sideways. The tunnel had a half-liner design, and the working part was 114 mm wide and 86 mm high. Some flow diagnostic methods were:

Effect of Corner Suction

Corner suction was applied through tangential slots at three streamwise locations ($x_{slot} = 20, 100, 150$ mm). It was observed that suction near or downstream of the interaction (e.g., $x_{slot} = 100$ mm) reduced corner separation by up to 27%.

Effect of Micro-Vortex Generators (VGs)

Two vane-type micro-VGs (3 mm high) were placed at each corner at x vg = 50mm. These dramatically reduced corner separations (up to 45%) while somewhat increasing centerline spanwise separation, which means that the flow field was more two-dimensional. The best results came from using both micro-VGs atx vg = 50mm and corner suction atx slot = 100 mm). The corner separation was cut down by 65%, and the centerline separation became broader but more even, which made the three-dimensionality much less noticeable.

Table 9: Separation parameters for different flow control configurations

Case	**Centerline (Span)**	**Corner (Left)**	**Corner (Right)**
Baseline	60 mm	21 mm	23 mm
Suction at $x = 100$ mm	58 mm	16 mm	19 mm
Micro-VGs at $x = 50$ mm	64 mm	12 mm	13 mm
VG + Suction	**70 mm**	**8 mm**	**8 mm**

2 OBJECTIVE OF WORK

The primary objective of this study is:

> *To investigate the properties of Shock-Boundary Layer Interaction (SBLI) due to shock impingement on a UDF-driven oscillating Inlet wall (2D) using Reynolds-Averaged Navier-Stokes (RANS) simulations.*

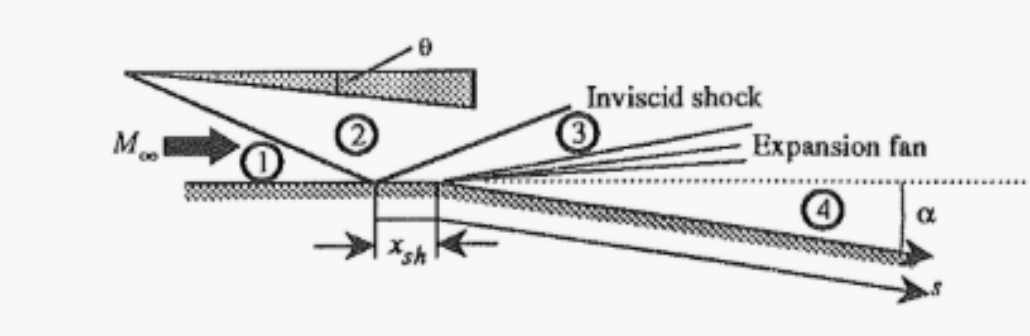


Figure 3: Schematic of the SBLI setup showing shock impingement on an expansion ramp.

Figure 3 illustrates the physical mechanism being studied. A freestream Mach number M_∞ encounters a wedge of angle θ, producing an oblique shock that impinges on a downstream ramp of angle α. The impingement point, denoted as x_{sh}, plays a critical role in the nature of the SBLI. As the ramp oscillates in time, x_{sh} varies dynamically, altering the local interaction structure between the shock and boundary layer.

The study aims to understand how this time-varying shock impingement location affects:

- Shock motion and strength near the expansion corner
- Boundary-layer separation or reattachment
- Pressure fluctuations and unsteadiness in the domain
- Effectiveness of turbulence modeling in capturing transient SBLI

By using RANS simulations with the Realizable k-ε turbulence model, the flow physics of these interactions are analyzed in detail to improve understanding of SBLI mechanisms in unsteady environments.

3 Computational Setup

3.1 Governing Equations

The compressible, unsteady turbulent flow is modeled using the Reynolds-Averaged Navier–Stokes (RANS) equations, which represent time-averaged conservation laws for mass, momentum, and energy. The system of equations is closed using the realizable k–ε turbulence model. The governing equations are:

- **Continuity (Mass Conservation):**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- **Momentum Equation:**

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \left[\mu_{\text{eff}} \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T \right) \right]$$

- **Energy Equation:**

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot [(\rho E + p)\mathbf{u}] = \nabla \cdot (k_{\text{eff}} \nabla T) + \Phi$$

Here, μ_{eff} is the effective dynamic viscosity including turbulent contributions, and Φ is the viscous dissipation function.

3.2 Turbulence Modeling: Realizable k–ε Model

The realizable k–ε model is a two-equation eddy viscosity model that solves transport equations for the turbulent kinetic energy (k) and its dissipation rate (ε). The model includes a new formulation for turbulent viscosity and a modified dissipation equation to ensure realizability.

- **Turbulent Kinetic Energy (k):**

$$\frac{\partial (\rho k)}{\partial t} + \nabla \cdot (\rho k \mathbf{u}) = P_k - \rho \varepsilon + \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right]$$

- **Dissipation Rate (ε):**

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \nabla \cdot (\rho\varepsilon \mathbf{u}) = C_1 S\varepsilon - C_2\rho\frac{\varepsilon^2}{k+\sqrt{\nu\varepsilon}} + \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\nabla\varepsilon\right]$$

- **Turbulent Viscosity:**

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad \text{with} \quad C_\mu = \frac{1}{A_0 + A_s\frac{U^* k}{\varepsilon}}$$

3.3 Computational Domain and Grid

The simulation domain is represented as a **2D planar** Inlet featuring an expansion ramp (Inlet wall), where SBLI occurs as a result of shock impingement. We looked into two shapes: one with a constant ramp and one with a ramp that changed over time based on a UDF. The following is a summary of mesh characteristics:

Mesh characteristics are summarized below:

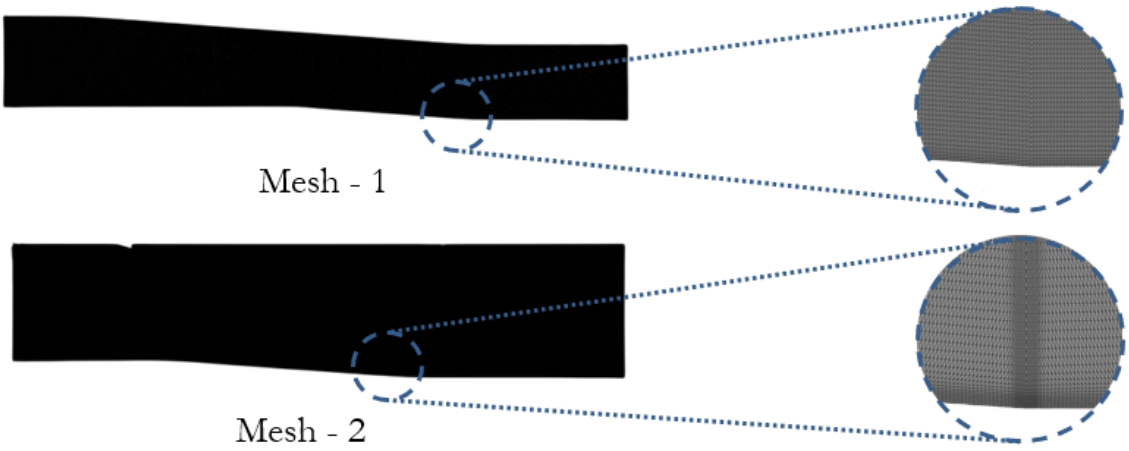


Figure 4: Mesh 1 & Mesh 2

Table 10: Mesh Characteristics

Feature	**Mesh–1**	**Mesh–2**
Element Size	2 mm	1 mm
No. of Elements	49134	335996
No. of Nodes	96432	169392
Tools Used	Face Meshing, Edge Sizing	Face Meshing, Edge Sizing, Vertex Sizing
Methods	Quad Dominant, Sweep	Quad Dominant, Multizone

3.4 Numerical Scheme: Advection Upstream Splitting Method (AUSM)

The AUSM (Advection Upstream Splitting Method) scheme is employed for inviscid flux calculation. It splits the flux into convective and pressure parts:

$$\mathbf{F}_{\text{total}} = \mathbf{F}^{(c)} + \mathbf{F}^{(p)}$$

- Convective flux is handled using an upwind-biased Mach number:

$$m_{1/2} = \rho a M_{1/2}, \quad \text{with} \quad M_{1/2} = M_L^+ + M_R^-$$

- Pressure flux is split as:

$$p_{1/2} = p^+(M_L) + p^-(M_R)$$

- AUSM is good for compressible flows since it can handle shocks and is strong in all Mach number ranges.

This method is ideal for SBLI problems due to its ability to sharply resolve shocks and expansion fans without excessive numerical dissipation.

3.5 Boundary Conditions

The boundary conditions applied are:

Table 11: Boundary Conditions

Inlet	M = 2.5; V = 867.98 m/s
Outlet	Guage Pressure = 0
Walls	No slip condition
Inlet Wall	Geom-1: Fixed ramp surface; Geom-2: Oscillating ramp with an amplitude of 4 Degree

4 Results & Discussion

This section shows the results of the unstable RANS simulation done for Geometry 1, which has a UDF-driven oscillating ramp. We provide a thorough study of the compressible flow field using both qualitative and quantitative visualizations, such as numerical schlieren, Mach number contours, shock strength changes along the centerline, and **FFT** of **unsteady pressure** data. The simulation shows how **shock–boundary layer interaction (SBLI)**, wave propagation, and pressure unsteadiness happen while the ramp moves.

4.1 Numerical Schlieren Visualization



Figure 5: Numerical Schlieren visualization at an instant during ramp oscillation.

Figure 5 displays the numerically calculated schlieren field derived from density gradients. The crisp, bright bands show severe oblique shocks, and the smoother, divergent areas show expansion fans. Notable characteristics seen:

- A powerful oblique shock from the upstream wedge stays steady during the ramp motion, which shows that there is no upstream feedback since the SBLI is weak.

- There is a dense network of shock and expansion waves near the ramp corner and downstream. This shows that the ramp is causing dynamic interactions.
- The wave reflection and junction become more complicated near the outlet, which captures temporary aspects of the wall that is oscillating.

4.2 Mach Number Contours over Time

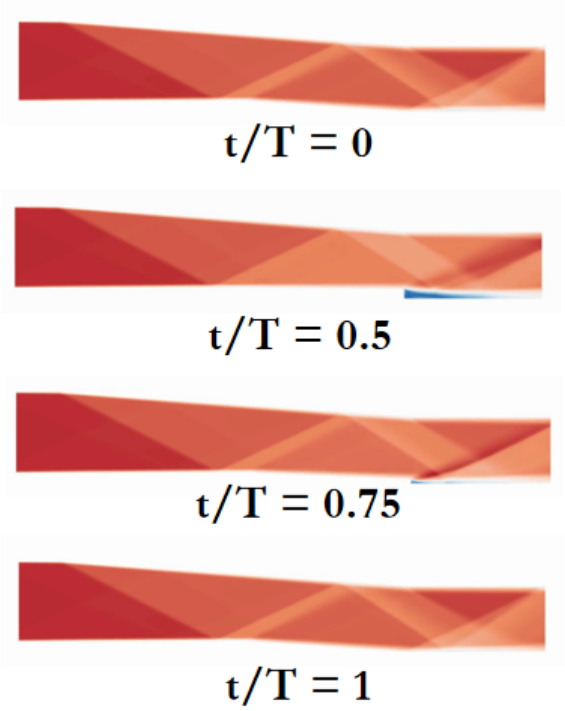


Figure 6: Mach number contours at four ramp oscillation phases: $t/T = 0,\ 0.5,\ 0.75,\ 1$.

Figure 6 displays the time-sequenced Mach contours over one ramp oscillation cycle. The ramp begins at its neutral position ($t/T = 0$), oscillates upward ($t/T = 0.5$), returns ($t/T = 0.75$), and resets to its initial state ($t/T = 1$). Notable trends include:

- The upstream oblique shock remains unaffected, supporting the claim of weak SBLI and no upstream pressure propagation.
- As the ramp deflects upward, the shock attached to the ramp intensifies and moves upstream. Expansion fans grow sharper, modifying the post-shock flow.

- At $t/T = 0.75$, the wave system partially returns to the neutral state, though minor asymmetries remain.
- The contour at $t/T = 1$ shows a near-complete reset of the wave structures, validating periodicity.

4.3 Pressure-Based Shock Strength Variation

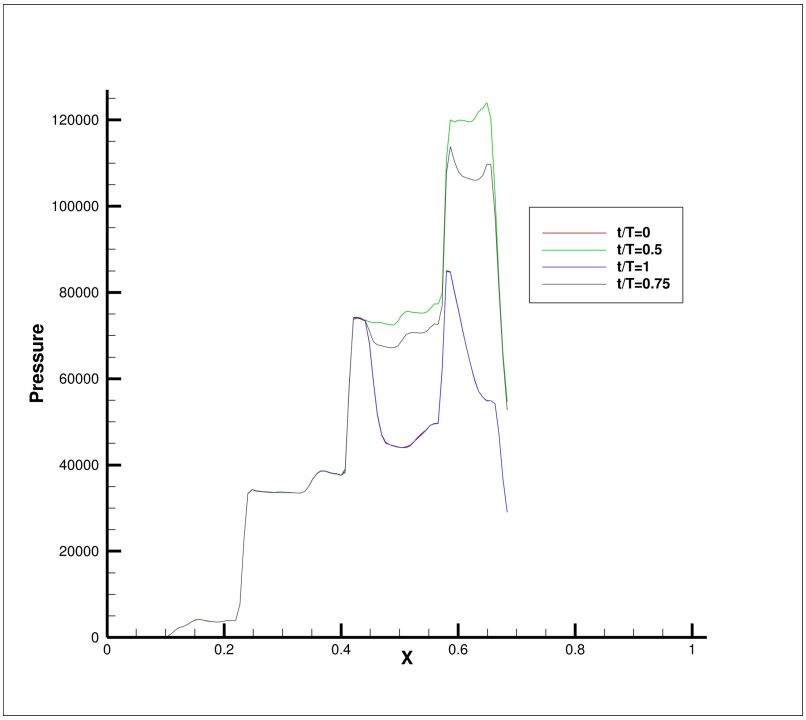


Figure 7: Centerline pressure distribution across the oscillation cycle (left) and corresponding Mach contours (right).

The central line pressure variation in Figure 7 highlights the dynamic response of shocks to ramp motion:

- The initial shock (x 0.25) remains unchanged in strength and location throughout the cycle, due to lack of upstream feedback.
- Shocks closer to the ramp (x 0.5–0.7) exhibit strong temporal variation. Their strength increases at $t/T = 0.5$ the ramp deflected upward and relaxes by $t/T = 0.75$.

- At $t/T = 1$, shock strength and spacing return to baseline, completing one full oscillation cycle.

This confirms that ramp motion predominantly affects downstream shock structures while leaving upstream wave features unaffected.

4.4 FFT of Unsteady Wall Pressure Data

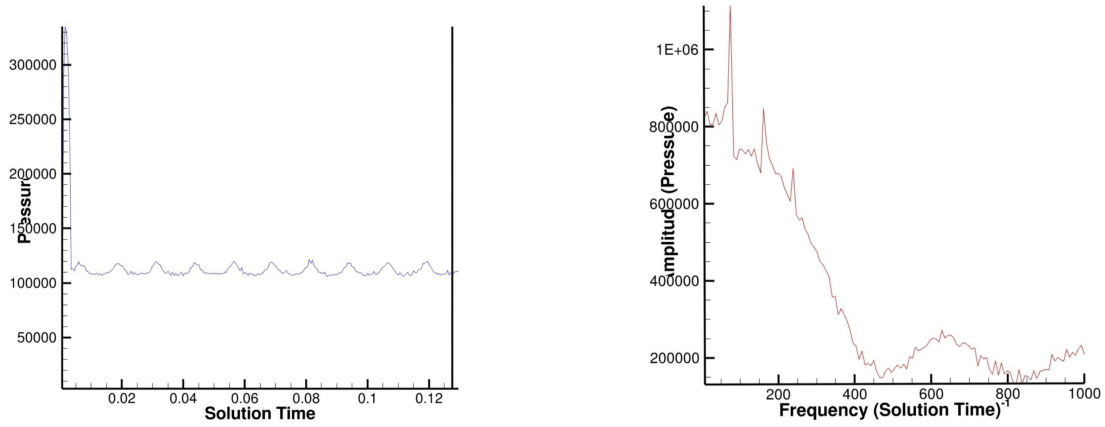


Figure 8: Left: Unsteady pressure signal at wedge wall. Right: Frequency spectrum (FFT) of pressure data.

The unsteady pressure signal recorded at the wedge wall is shown in Figure 8 (left). The waveform exhibits small but periodic fluctuations due to the influence of downstream wave activity.

The corresponding FFT spectrum (Figure 8, right) reveals:

- A dominant low-frequency peak matches the oscillation frequency of the ramp.
- Minor harmonic peaks indicate nonlinear interactions or secondary wave reflections.
- Absence of broadband high-frequency content confirms the lack of turbulence or separation bubbles upstream.

This validates that even in the absence of strong upstream SBLI, the unsteady pressure is subtly modulated by downstream flow dynamics.

5 Challenges Faced

1. **Development of UDF:** To define the amplitude of the divergence angle, the casefile has to run multiple times to check the safe limit of the mesh movement.

2. **Failed Mesh Movement:** During the movement of the Inlet wall, the mesh must adjust in such a way that each element has a finite volume after the stretching, but the mesh movement failed multiple times due to the negative volume of the elements.

3. **Adjustment of the Divergence Angle:** The amplitude of the divergence angle should be very small for optimal mesh movement; otherwise, the mesh movement will fail due to excessive stretching.

4. **Meshing Error:** The simulation stopped multiple times due to defeatured elements in the mesh. Sometimes fluent failed to initialize the calculations due to an unstructured mesh full of triangular elements.

5. **Total Run Time:** As it was a transient simulation, it took almost 2 days to converge for each case. It diverged multiple times as the CFL number crossed 0.6.

6 CONCLUSION

The numerical investigation successfully captured the unsteady flow features associated with the 2D ramp-wall motion using the realizable $k - \epsilon$ model.

1. The ramp motion did not affect the upstream oblique shock created by the wedge, which suggests that the SBLI regime is weak and has little feedback from upstream.

2. In the downstream region, on the other hand, there were strong interactions between shock and expansion waves. The strength and structure of these contacts changed over time with the ramp oscillation.

3. Mach contour sequences proved the flow field's periodic unstable behavior by revealing that the ramp phase angle had a big effect on the shock position and strength.

4. The pressure distribution along the centerline corroborated the periodicity and repeatability of the dynamic event, showing clear changes in shock strength over time near the expansion corner before returning to baseline after one oscillation cycle.

5. The ramp oscillation frequency was identified as the dominating low-frequency component in the FFT analysis of wall pressure changes. with minor higher-order harmonics caused by secondary reflections. The absence of broadband high-frequency energy confirmed the lack of upstream separation or turbulent bursting.

6. The AUSM flux scheme effectively resolved the shock structures and captured the unsteady wave dynamics without introducing significant numerical dissipation.

References

[1] Y.Guo, H.Tan, Y.Zhang, X.Li, H.Xue, Z.Luo, and H.Huang, "On incident shock wave/boundary layer interactions under the influence of expansion corner," *Journal of Fluid Mechanics*, vol. 988, p. A28, 2024.

[2] X.Xiang and H.Babinsky, "Corner effects for oblique shock wave/turbulent boundary layer interactions in rectangular channels," *Journal of Fluid Mechanics*, vol. 862, pp. 1060–1083, 2019.

[3] X.Xiang and H.Babinsky, "An Experimental Study of Corner Flow Control Applied to an Oblique Shock-Wave/Boundary-Layer Interaction," in *2018 AIAA Aerospace Sciences Meeting*, AIAA Paper 2018-1532, 2018.

[4] Y.T. Chew, "Shockwave and boundary layer interaction in the presence of an expansion corner," *Aeronautical Quarterly*, vol. 30, no. 3, pp. 506–527, 1979.

[5] Y.T. Chew and L.C. Squire, *The Boundary Layer Development Downstream of a Shock Interaction at an Expansion Corner*, Aeronautical Research Council R&M 839, 1978.

[6] F.K. Lu and K.-M. Chung, "Downstream Influence Scaling of Turbulent Flow past Expansion Corners," *AIAA Journal*, vol.30, no.12, pp.2976–2977, 1992.

[7] K.-M. Chung and F.K. Lu, "Hypersonic Turbulent Expansion-Corner Flow with Shock Impingement," *Journal of Propulsion and Power*, vol. 11, no. 3, pp. 441–447, 1995.